SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5112

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 10%

QUESTIONS: 5 **Total Marks:50**

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I ___Bhavya PS___ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

TECHNICAL PRESENTATION

1. From Caesar to Vigenère: Evolution of Substitution Ciphers”
2. “Substitution vs Transposition: Which Provides Better Security?”
3. “Security Attacks, Threats, Risks and Their Impact on Security Goals”
4. “Steganography vs Cryptography: Hidden Security in the Digital World”
5. “Role of Number Theory in Cryptography: From Euclid to Euler”

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Substitution vs Transposition: Which Provides Better Security?

CSA5112 CRYPTOGRAPHY AND NETWORK SECURITY [SLOT – B]

Presented by
Bhavya ps

Introduction

- Classical cryptography mainly uses two techniques to hide information:
- **Substitution Cipher** – Replace characters with other characters.
- **Transposition Cipher** – Rearrange the positions of characters.
- Both aim to protect data, but they do so in different ways.
- **Image Idea:**
  🔒 → ABC → ??? → Secure Message

What is a Substitution Cipher?

Definition
A **Substitution Cipher** replaces each letter of the plaintext with another letter, number, or symbol according to a fixed rule.

Simple Example (Caesar Cipher)
Plaintext: HELLO (**K = 3)**

Result:
HELLO
↓
KHOOR

What is a Transposition Cipher?

Definition

A **Transposition Cipher** keeps all the original letters but changes their positions.

Example

Plaintext: HELLO

(Reverse the letters)

HELLO

↓↓↓↓↓

OLLEH

Which is More Secure?

Let's Compare

Suppose the message is : ATTACKATDAWN

Substitution

DWWDFNDWGDZQ

<u>The positions stay the same.</u>

Transposition

ATNTAWTAKDAC

<u>The letters stay the same, but they are rearranged.</u>

If an attacker knows English...

Best Security: Combine Both

Modern encryption combines both techniques.

Example

Original : HELLO

Step 1 – Substitution (K=3)

HELLO

↓

KHOOR

Step 2 – Transposition (Reverse)

KHOOR

↓

ROOHK

THANK YOU!!